

ASEN 5090 Lecture 2

Navigation Measurements & Methods

TODAY

- Definitions
- Brief overview of navigation sensors/systems
- Basic models for range, pseudorange, Doppler
- Basic models for trilateration, hyperbolic positioning, and Doppler positioning
- Reading – ME Ch1
- HW1 posted on Canvas – Due 9/4 11PM on Gradescope*
Allows for late submissions until 9/6, but ANY late submission counts against allocation of 2

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Navigation & Related Definitions

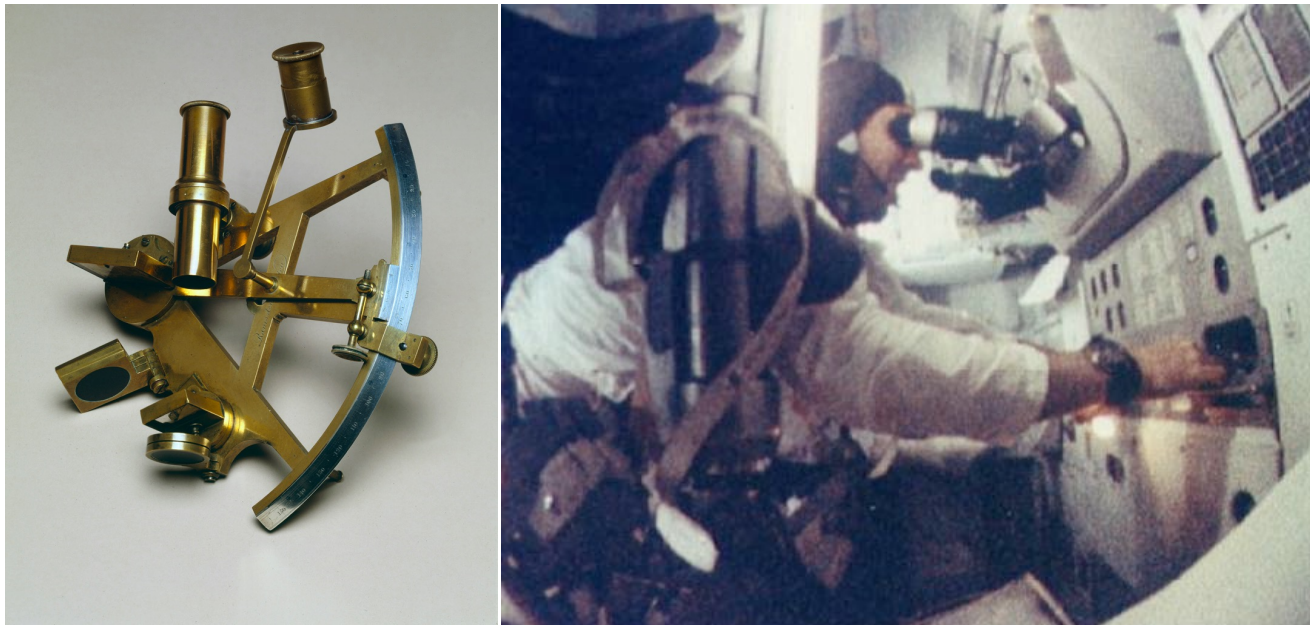
- **Geodesy** – what is the size, shape, and gravity field of the Earth?
- **Positioning** – where is/was it?
- **Navigation** – where am I now?
- **Guidance** – which path leads to my destination?
*Doesn't necessarily tell you where you are,
just indicates which way to go: landing beacons, ILS, MLS*
- **Control** – how can I make my vehicle follow the desired path?

Landmark & Dead reckoning

- **Landmark** – something you recognize that can be related to a map.
Triangulation is based on angular measurements to landmarks.
Ranging is based on distance measurements to landmarks.
The problem is what to do when you can't see a landmark.
- **Dead reckoning** – knowing where you are starting from and keeping track of distance and direction traveled.
Usually can measure speed and direction, then keep track of time.
The problem is that errors tend to build up over time.
- **Earth & Maps** – fundamental to both types of positioning is a representation of the earth. Maps and astronomy developed together in ancient times.

CELESTIAL NAVIGATION

- **Celestial navigation** – measurement of altitude of stars or planets
- Measurement of celestial events allows for comparison of local times and hence determination of longitude. Required accurate ephemerides / predictions of the motions of the planets & moons, led to observatories, and was favored by Newton and Halley



Images from the ION Virtual Museum :
<http://www.ion.org/museum/>
Left – Sextant, Right – Astronaut Alan Lovell using Apollo sextant for star sighting

INERTIAL NAVIGATION

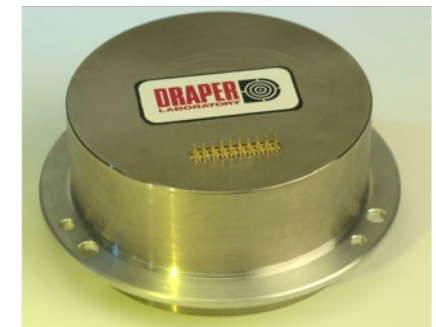
- A set of 3 accelerometers on a stable platform measure specific force acting in each of 3 orthogonal directions. Removing gravity allows the measurements to be integrated to get velocity and again to get position.
- Accurate compensation for gravity in the integration is critical – unstable in vertical direction.
- Requires a set of gyroscopes (gyros) to keep the platform stable or isolated, or keep track of how it is changing (strapped down INS).
- INS is a dead reckoning system. Drifts in accelerometers and gyros lead to errors, so INS requires occasional (or frequent) position fixes.



Draper SPIRE IMU Gimbal System

<https://www.eeworldonline.com/inertial-measurement-units-the-hidden-key-to-apollo-success-now-a-mems-device/>

<https://arc.aiaa.org/doi/pdf/10.2514/3.19748>



8.1 in³ MEMS IMU



RADIO NAVIGATION

- The discovery of radio waves led to wireless communications, radar, and radio navigation
- Radio waves are classified by their frequency (wavelength = $c/\text{frequency}$)
 $c = 299,792,458 \text{ m/s}$ (exact value)

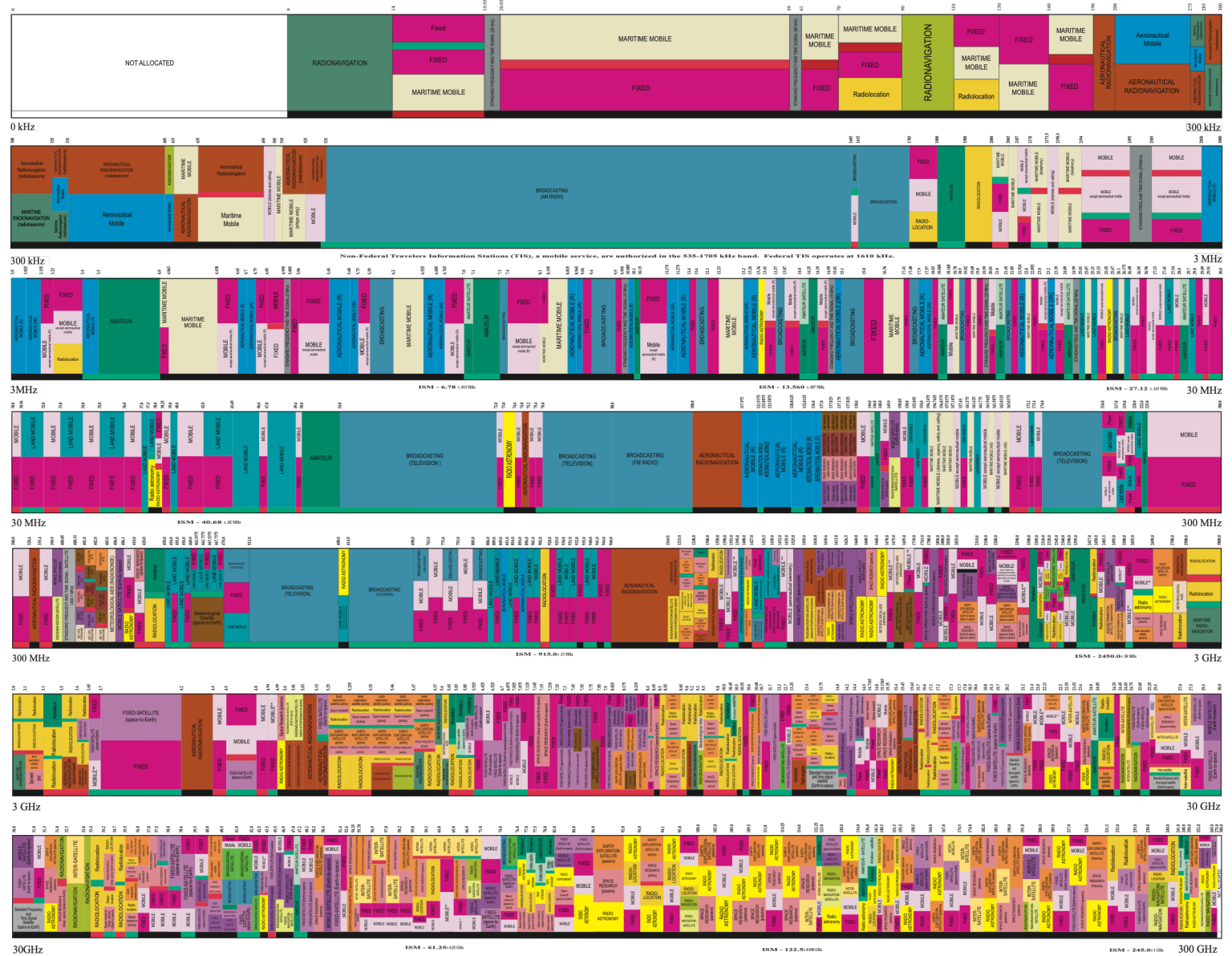
- VLF - < 30 kHz
- LF 30-300 kHz
- MF 300 kHz – 3 MHz

- HF – 3-30 MHz
- VHF – 30-300 MHz
- UHF – 300 MHz – 3 GHz (GPS is here) (According to IEEE only goes to 1 GHz)



UNITED STATES FREQUENCY ALLOCATIONS

THE RADIO SPECTRUM



PLEASE NOTE: THE SPACING ALLOTTED THE SERVICES IN THE SPECTRUM SEGMENTS SHOWN IS NOT PROPORTIONAL TO THE ACTUAL AMOUNT OF SPECTRUM OCCUPIED.

https://www.ntia.doc.gov/files/ntia/publications/january_2016_spectrum_wall_chart.pdf

RADIO WAVE PROPAGATION

- Radio signals can be used for navigation if you know the propagation path.
- Ground wave - Low frequency radionav VLF – MF propagate by ground wave – this is useful for beyond the horizon. Ground based radio nav such as Loran and Omega use this.
- Sky wave - HF signals travel some along the ground, some reflect off the ionosphere, some go straight – sky wave
- Direct line of sight - VHF and higher only travel in straight lines

LORAN – Long range navigation system

- Developed 1940's by US as a military navigation aid. (others DECCA, Chayka and Gee.)
- Ground-based hyperbolic system for marine and aviation.
- LF band – 90-110 kHz

LORAN-C

- Converted to civil marine use in 1974 and was operated by the USCG until 2010.
- Initially provided coverage for maritime navigation in U.S. coastal areas, but eventually served continental states, coastal areas, and parts of Alaska.
- Can be used for 2D aircraft and ship navigation.
- LORAN-C was approved as an en route supplemental air navigation system for both Instrument Flight Rule (IFR) and Visual Flight Rule (VFR) operations.
- eLORAN – Enhanced LORAN-C includes transmissions synchronized to UTC, Differential Corrections, & other improvements. Ongoing consideration of eLORAN as a GPS backup.

See PNT2021 Ch 41 for details

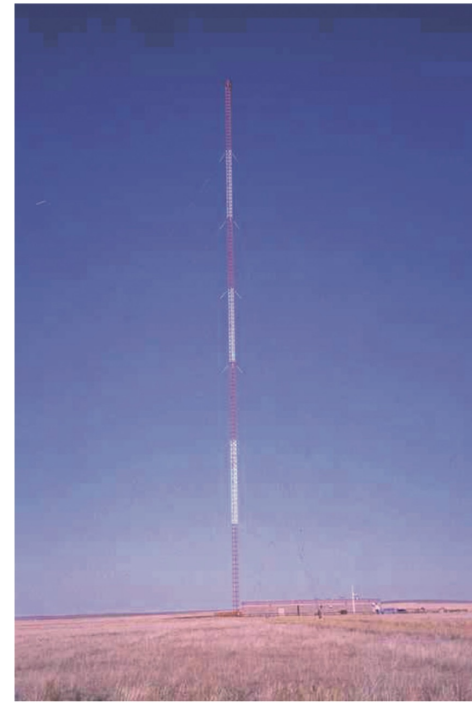


Figure 41.7 700-ft TLM Loran antenna. Source: UrsaNav Inc. Photo.

Loran-C Chains & Signal paths

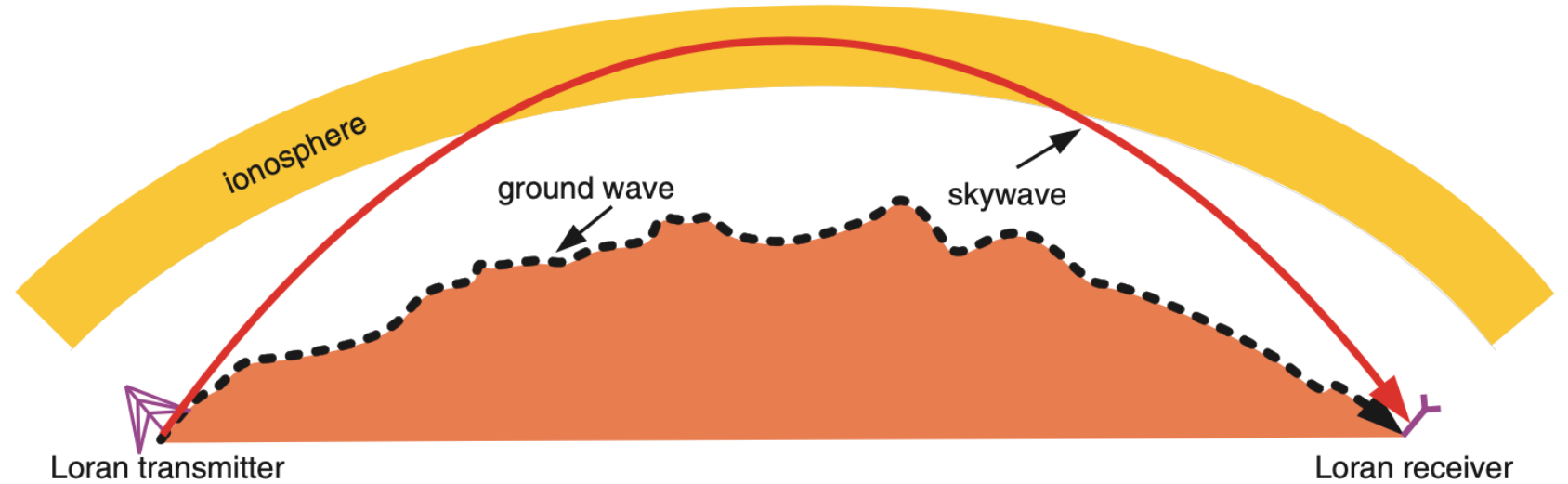
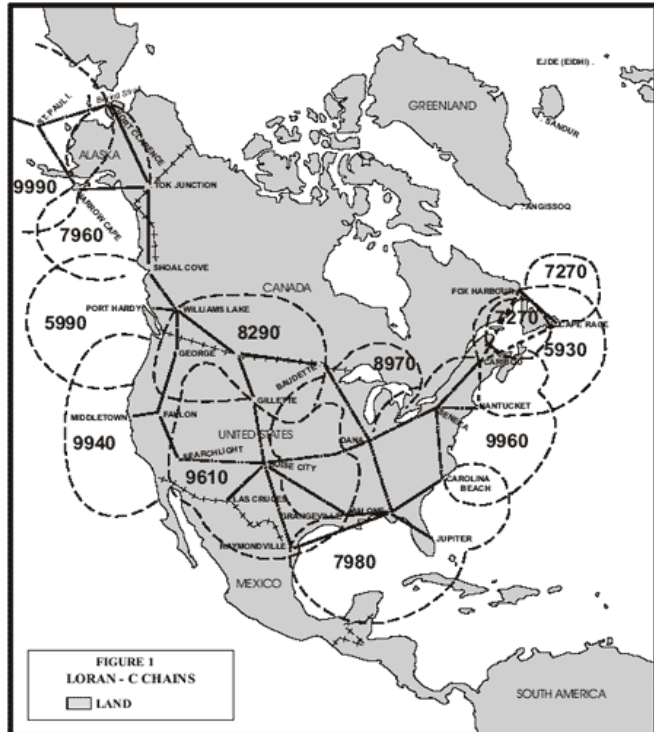


Figure 41.9 Loran-C ground wave and sky-wave propagation. *Source:* Illustration courtesy of Wouter Pelgrim, C

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Navigation with Low-Frequency Radio Signals

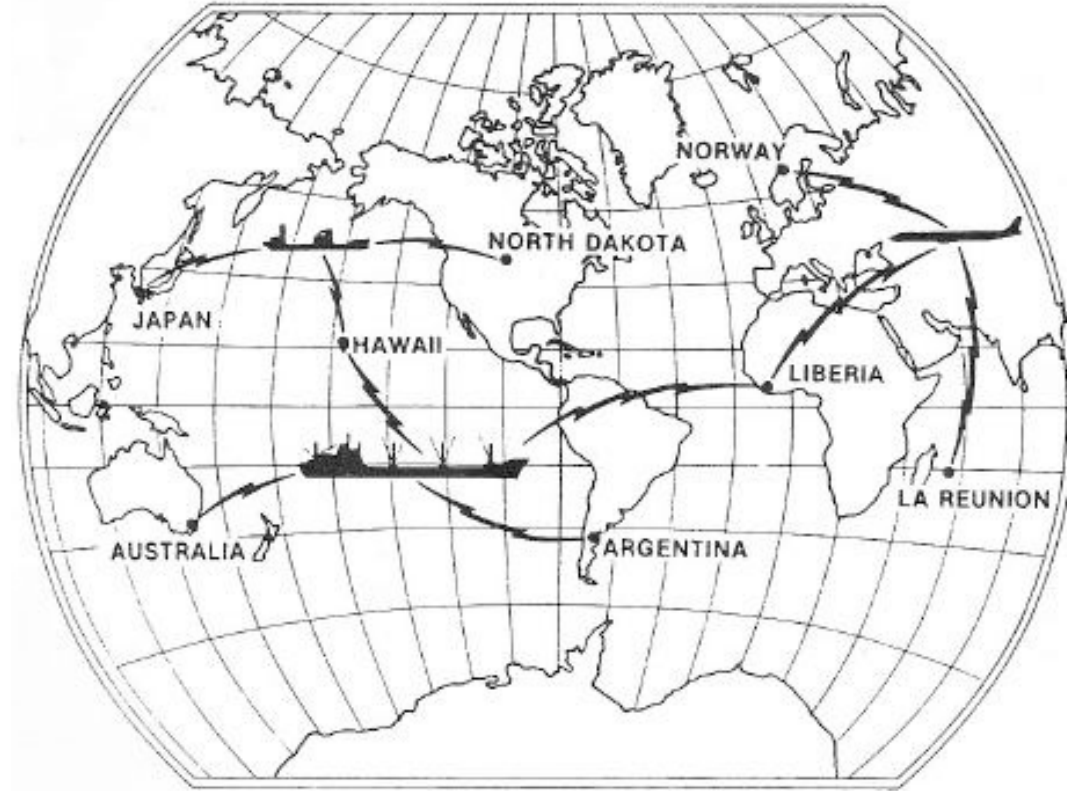
Wouter Pelgrim¹ and Charles Schue, III²

¹ Blue Origin LLC, United States

² UrsaNav, Inc., United States

OMEGA – First global radio navigation system

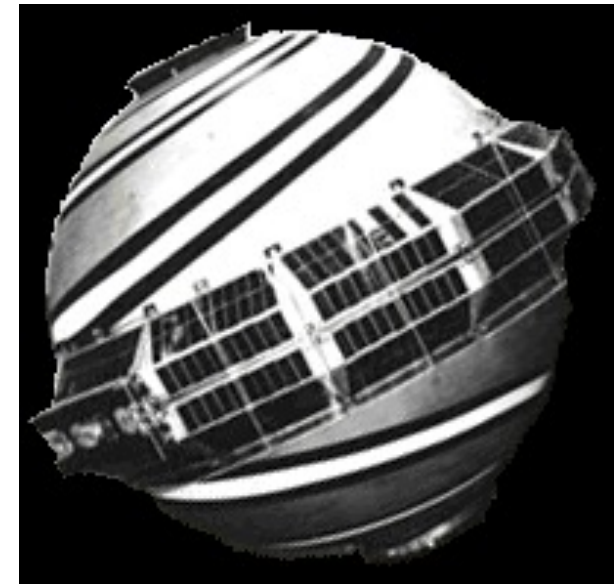
- Became operational in early 1970's. VLF radionav - 10-14 kHz
- Transmitters broadcast 10 kW of power from antenna towers 366-457m or "valley spans" up to 3500m
- 8 stations around the world
- Used for shipping & aircraft
- FDMA – each station transmitted 4 common frequency (CW) tones + one unique freq
- Receiver measured phase difference between transmitters and resolved ambiguities in lanes.
- Accuracy ~ few km
- Decommissioned in 1997



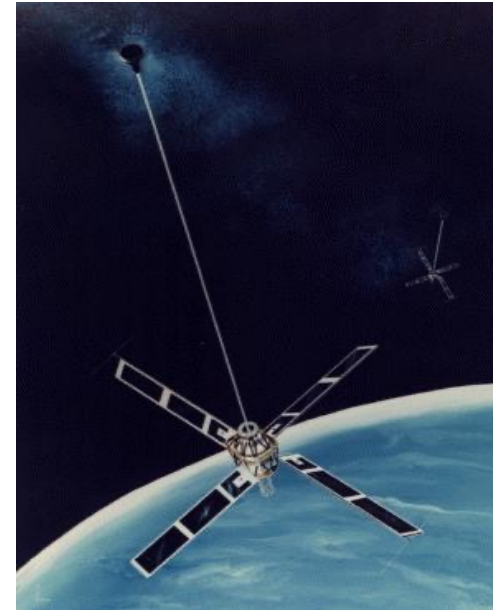
<http://www.auroralchorus.com/omega.htm>

TRANSIT – First navigation satellite system

- Late 1950's after Sputnik, APL at the Johns Hopkins University developed the system based on Doppler positioning. Sponsored by the Navy to provide position fixes & reset INS for submarines.
- There were 4-7 satellites in LEO polar orbits (90 minute period). One satellite in view at a time.
- Transmit at 150 MHz and 400 MHz
Measurement of Doppler and rate of change of Doppler + knowledge that you are on the sea surface pinned it.
- Requires user motion to be slow or very well known.



From Smithsonian website:
<http://www.nasm.si.edu/exhibitions/gps/before.html>



Radio Navigation Measurements

- **Trilateration** - time of arrival

- If signal velocity is known, range from the transmitter to receiver is measured by $v^*(t_{rec} - t_{trans})$.
- Position is determined by intersection of circular lines of positions centered on the transmitters.
- Must assume perfect clock for this work.

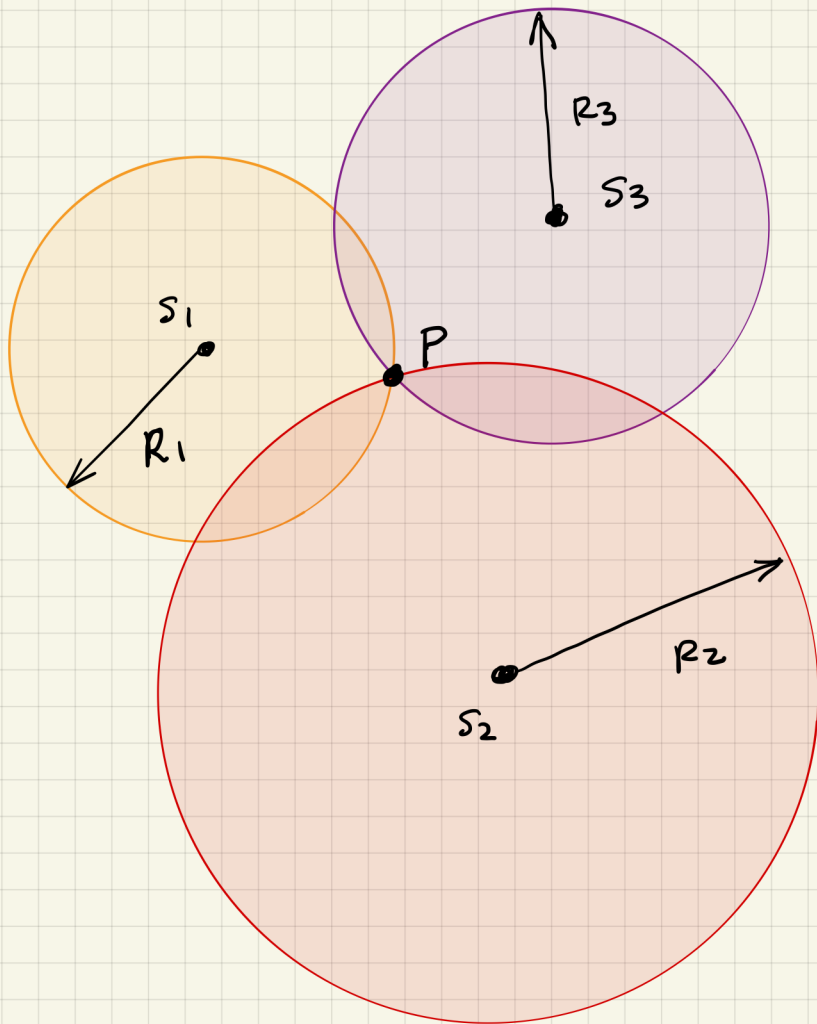
- **Hyperbolic positioning** - time difference of arrival

- Receiver measures the difference in arrival time of signals from two synchronized transmitters.
- Position is determined by intersection of hyperbolic lines with foci at the transmitter pairs.

- **Doppler positioning** - frequency shift

- As the transmitter approaches the frequency is shifted higher due to Doppler effect (relativity).
- At the point of closest approach the shift is zero, and it goes lower as the transmitter moves away.
- If the trajectory of the transmitter is known, you can get 1D or 2D position information from this measurement and its rate of change.

TRILATERATION



IDEAL

$$R_1 = \sqrt{(x_1 - x)^2 + (y_1 - y)^2}$$

$$R_2 = \sqrt{(x_2 - x)^2 + (y_2 - y)^2}$$

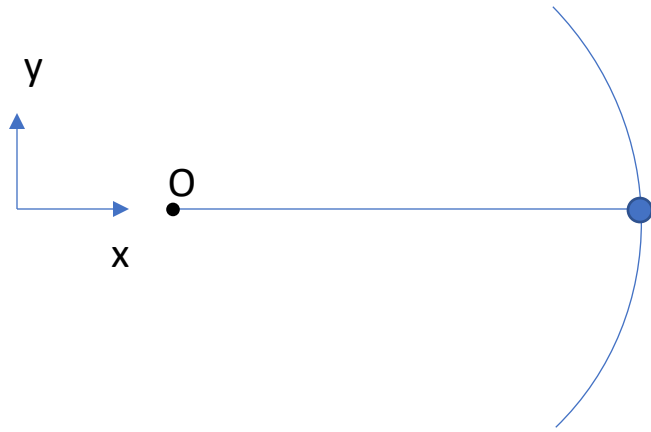
$$R_3 = \sqrt{(x_3 - x)^2 + (y_3 - y)^2}$$

Range or TOA time of arrival

Questions:

1. What assumptions are made in the range equation?
2. How should we solve this?
3. What factors affect the accuracy of the solution?
4. If you have more than two circles, will they intersect at one location?

Range Measurement Sensitivity - *what makes a measurement useful?*



$$R^2 = (x_1 - x)^2 + (y_1 - y)^2$$

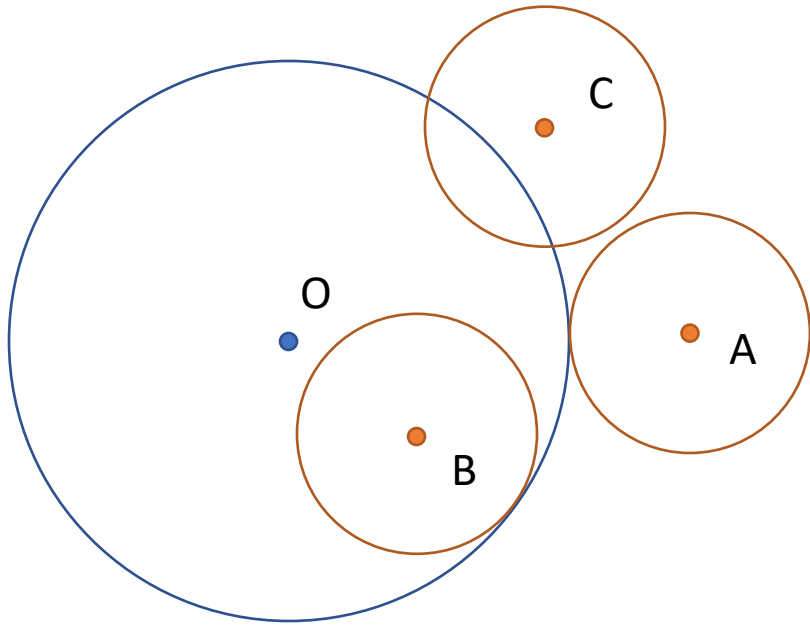
$$2R\delta R = 2(x_1 - x)(-\delta x) + 2(y_1 - y)(-\delta y)$$

$$\delta R = -\frac{x_1 - x}{R} \delta x - \frac{y_1 - y}{R} \delta y$$

- Range gives
 - Good information in radial direction
 - Poor information in tangential direction
- If you evaluate this at your location (surface of Earth), it tells the sensitivity of (δR) with respect to $(\delta x, \delta y)$

What measurement geometry leads to good position solutions?

- If you have a range measurement to O, which do you pick for your next measurement?



Pseudorange – time of arrival with imperfect user clock

...for now we ignore transmitter clock effects

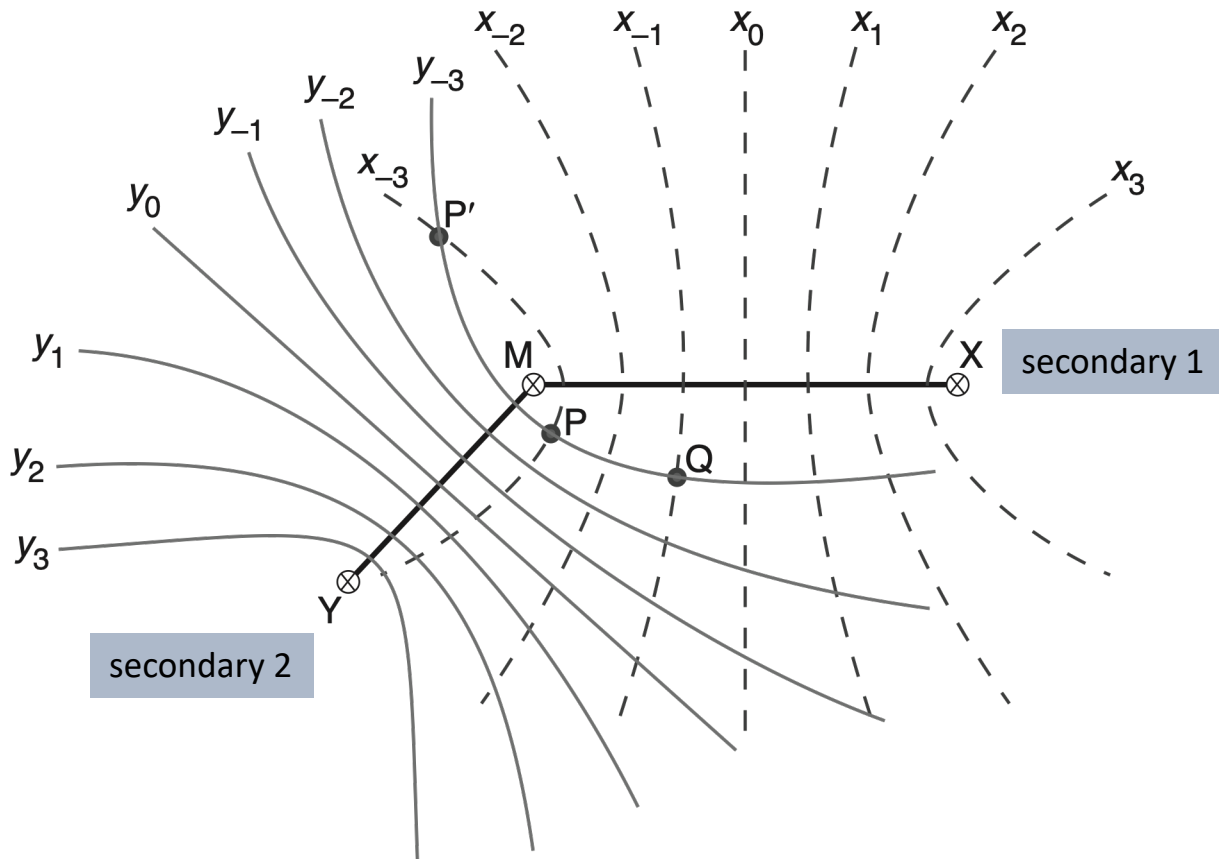
What do the circles look like now?

$$\rho_i = \sqrt{(x_i - x)^2 + (y_i - y)^2} + b$$

Do not use equation in ME Figure 1.9

+ b means that user clock is fast (later in time that it should be)

Hyperbolic Positioning in 2D



Measurement is the range difference between two transmitters at known locations; or time difference of arrival (TDOA)

$$r_0 = \text{range to } M$$

$$d_1 = r_0 - r_1$$

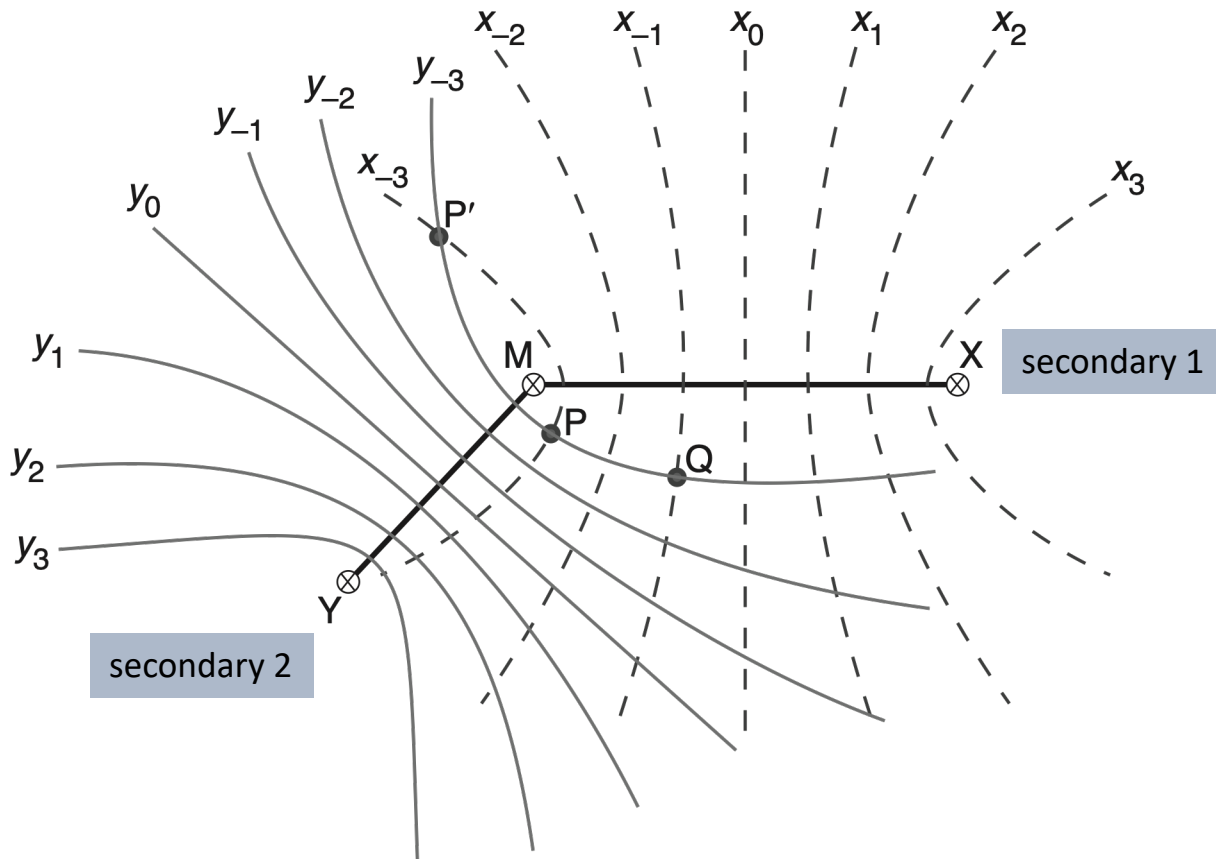
$$d_2 = r_0 - r_2$$

Lines show constant values of d .

What is the largest value of d ? Where?

Advantage over ranging is that user clock bias does not influence result.

Hyperbolic Positioning in 2D



- In Loran, S receives M, then transmits, as a way to synchronize the transmissions. The known delay due to separation is subtracted from observation.
- In eLoran, the stations are synchronized externally to a common time (UTC).

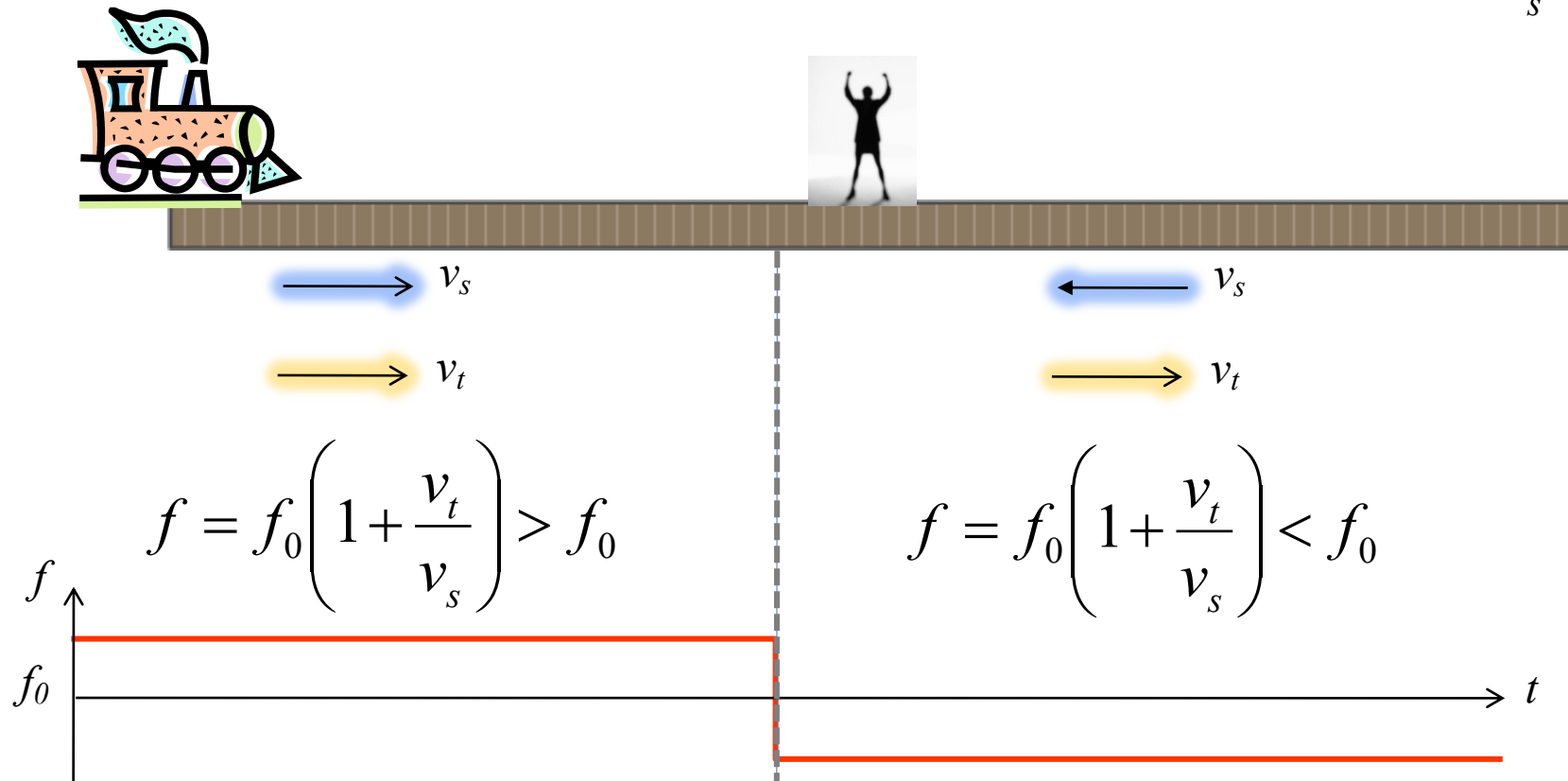
Doppler Positioning: Stationary Observer

$$f = f_0 \left(1 + \frac{v_t}{v_s} \right)$$

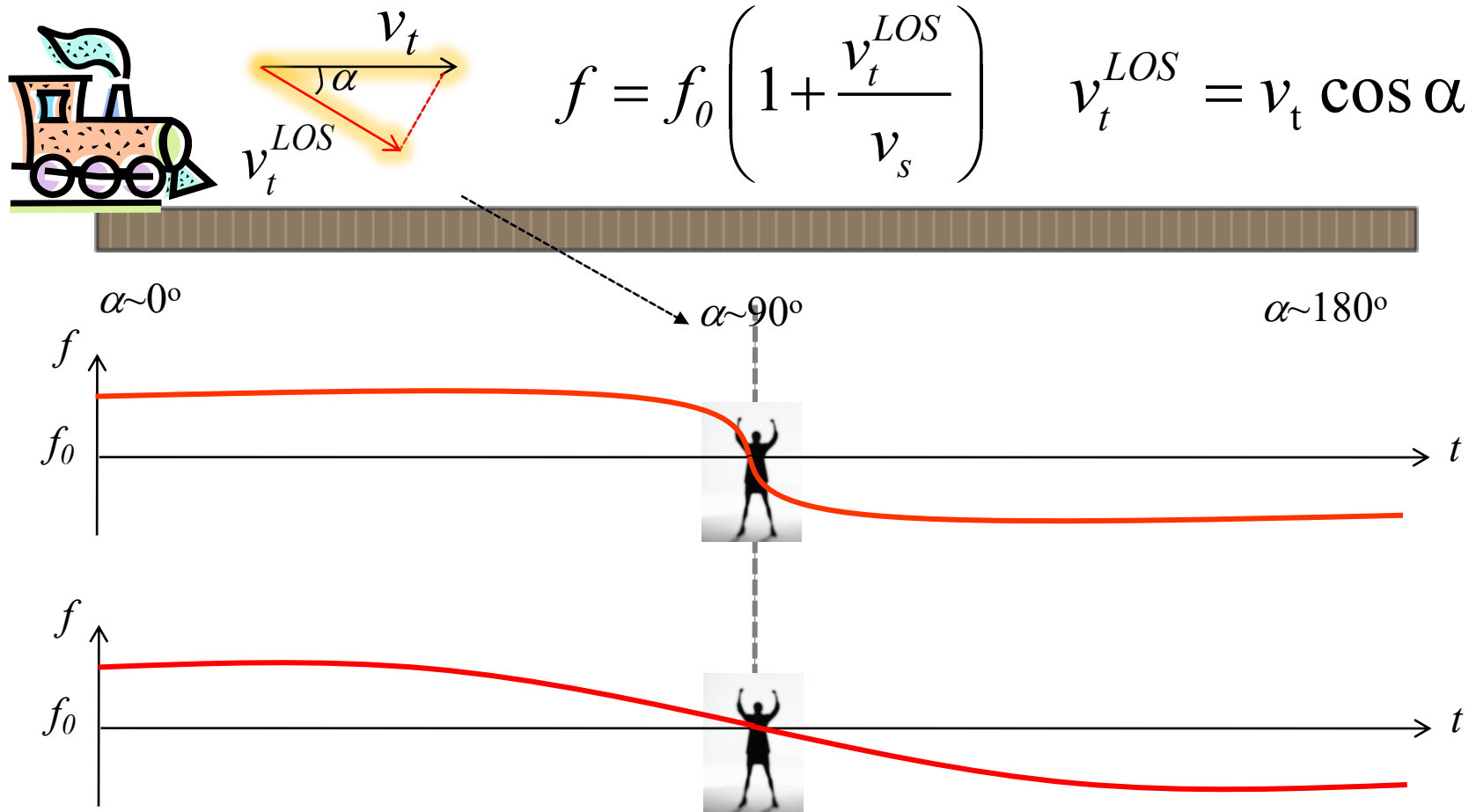
$v_s = \text{speed of signal}$

Doppler frequency:

$$f_D = f - f_0 = f_0 \frac{v_t}{v_s}$$

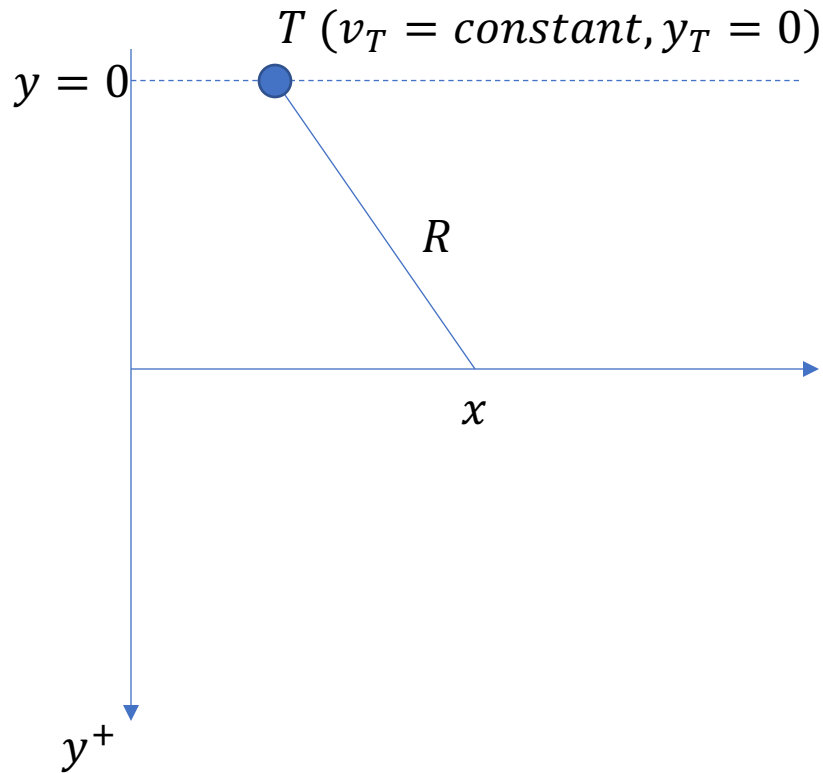


Doppler Positioning: How It Works (2)



Doppler Measurement

- If motion of the transmitter is known, we can observe a stationary receiver location



Simple example $R^2 = (x - x_T)^2 + y^2$

$$2R\dot{R} = -2(x - x_T) \dot{x}_T$$

$$\dot{R} = -\frac{x - x_T}{R} v_T = -\frac{x - x_T}{\sqrt{(x - x_T)^2 + y^2}} v_T$$

At closest approach ($\dot{R} = 0, R = y$)

$$\dot{R} = 0 \rightarrow x = x_T$$

$$\ddot{R} = \frac{v_T^2}{R} = \frac{v_T^2}{y}$$

$$y = \frac{v_T^2}{\ddot{R}_{max}}$$

Doppler Measurements for Radionavigation

$$\frac{f_r}{f_t} \approx \frac{1 - v_r \cdot \hat{e}/c}{1 - v_t \cdot \hat{e}/c} \approx 1 - v_r \cdot \hat{e}/c + v_t \cdot \hat{e}/c$$

$$\frac{f_r}{f_t} \approx 1 + \frac{v_t - v_r}{c} \cdot \hat{e}$$

$$f_D = -\frac{\dot{R}}{c} f_t \approx \frac{v_t - v_r}{c} \cdot \hat{e} f_t$$

Assume:

Signal travels at speed of light (c)

Clock frequencies are perfect

Speed of a satellite
in a circular orbit (radius, r)

$$v = \sqrt{\mu/r}$$

$$\mu_E = 398600.4 \text{ km}^3/\text{s}^2$$



Doppler Measurements for Signals of Opportunity Radionavigation

Received: 16 October 2020 | Revised: 20 June 2021 | Accepted: 5 July 2021

DOI: 10.1002/navi.438

ORIGINAL ARTICLE

WILEY 

Navigation using carrier Doppler shift from a LEO constellation: TRANSIT on steroids

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$$-\lambda D^j = \frac{(\hat{\rho}^j)^T(\vec{v} - \vec{v}^j) \left\{ \frac{1 + \frac{d\delta^j}{dT^j}}{1 + \frac{1}{c} [a_{\delta tp}^j - (\hat{\rho}^j)^T \vec{v}^j]} \right\} + c \frac{d\delta_R}{dT_R} - c \frac{d\delta^j}{dT^j}}{1 + \frac{d\delta_R}{dT_R}} \quad (7)$$

Abstract

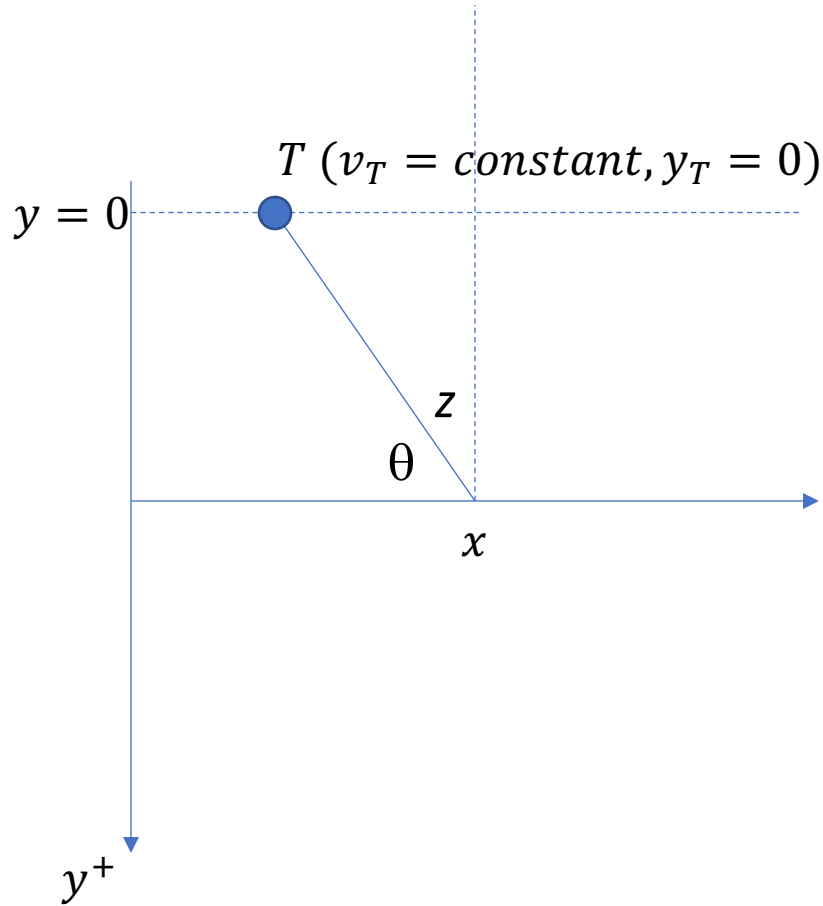
A new global navigation concept is studied that relies on carrier Doppler shift measurements from a large LEO constellation. This system could provide an alternative to pseudorange-based GNSS. The concept uses a high-fidelity model of received carrier Doppler shift. This model is used in a point-solution batch filter that simultaneously estimates eight unknowns: the three position vector components, receiver clock offset, three velocity vector components, and receiver clock offset rate. The filter uses eight or more measured Doppler shifts in its least-squares fit. A generalized Geometric Dilution of Precision (GDOP) analysis indicates that absolute position accuracies on the order of 1-5 meters and absolute velocity accuracies on the order of 0.01 m/sec to 0.05 m/sec may be achievable if the range-rate precision of the Doppler shift measurements is 0.01 m/sec. These accuracies are comparable to current pseudorange-based GNSS. Clock offset accuracy is on the order of 0.0001 to 0.0010 sec 1- σ .

KEYWORDS

batch filter, Doppler navigation, GDOP, large constellations



Angle Measurements with respect to a known position



Homework 1 – Posted & Due Friday 9/5 11 PM MDT

- Four problems
 1. 1-D pseudorange example
 2. Comparing utility of range, Doppler, and angle measurements from a ground station for determining location of an aircraft flying by
 - 3 & 4 are related to GPS signals

References

- Interesting article on Longitude in the Summer 2014 ION Newsletter: [*Ships, Clocks and Stars: The Quest for Longitude*](#)
- *From Sails to Satellites, The Origin and Development of Navigational Science*, by J. E. D. Williams, (available electronically via NetLibrary)
- The ION Virtual Museum: <http://www.ion.org/museum/>
- National Air & Space Museum: <http://www.nasm.si.edu/exhibitions/gps/>
- <http://www.aero.org/education/primers/gps/gpstimeline.html>
- <http://www.unoosa.org/oosa/SAP/gnss/icg/icg03/presentations.html>
- *Position, Navigation, and Timing Technologies in the 21st Century, Ed by Morton, et al., Vol 2, Ch 41 Pelgrum & Schue*

Acknowledgements

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